

SoloBand: A Maternity Support Band with the One-Handed Tightening System for Pregnant Individuals

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Abstract

Pregnancy induces substantial anatomical and biomechanical changes that often lead to discomfort such as lower back and pelvic pain. Maternity support garments have been shown to alleviate some of these symptoms by providing lumbar support and improving postural stability. However, current maternity support garments frequently fail to address the ergonomic and usability needs of pregnant individuals, particularly those requiring one-handed operation. This study aims to develop and evaluate *SoloBand*, a novel maternity support band that integrates a two-phase tightening system—thermally actuated Shape Memory Alloy (SMA) springs for initial self-anchoring, and a BOA® dial system for precise compression adjustment. Preliminary evaluations assessed three main design criteria: one-handed donning/doffing time, belly-lifting support, and compression force. Results demonstrated that users could secure the band in approximately 43 seconds using one hand, achieve an average abdominal lift of 0.59 inches (≈ 1.5 cm), and generate measurable compression forces up to 238 g at the lower back. The results of this preliminary evaluation support the feasibility of the *SoloBand* in achieving one-handed operability, highlighting its potential as a user-friendly solution for enhancing the usability of maternity support garments. Future work will focus on optimizing support and compression delivery, as well as developing a user-friendly and autonomous SMA activation system.

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1 Introduction

Pregnancy involves major hormonal, and biomechanical changes, including increased lumbar lordosis – typically around $18^\circ \pm 10^\circ$ in later trimesters [1-3]. These adaptations, often continuing postpartum, can cause lower back and pelvic pain [4]. Maternity support garments help relieve these symptoms by offering lumbar support and postural stability [5]. Postpartum, especially after Cesarean delivery, they assist in incision healing and abdominal support [6]. Compression and lift features can stabilize abdominal muscles weakened by diastasis recti [7]. However, many current designs overlook usability and ergonomic needs, revealing a gap in the maternity garment market.

Maternity support garments, including belts, briefs, cradles, and bodysuits, aim to enhance comfort and provide support for pregnant individuals and new mothers. Support belts are the most preferred for their ease of wear, adjustability, and convenience, yet most rely on straps and Velcro that require two hands. This design poses challenges for those with upper limb weakness, reduced grip strength, or pain, and for postpartum mothers often occupied with caring for a baby [1]. These limitations highlight the need for maternity belts that enable effective single-handed donning, doffing, and adjustment to improve usability.

In the field of Human-Computer Interaction (HCI), several wearable systems have been developed to support the needs of

pregnant individuals and new parents, particularly in health monitoring and stress reduction. For example, smart pregnancy journals that integrate wearable sensor data with self-tracking features have been shown to increase user awareness of maternal health, demonstrating above-average usability scores in user studies [8]. Wearable devices like smart wristbands have also been explored for monitoring physiological parameters – such as blood pressure and physical activity – to support the early detection of complications like gestational hypertension and diabetes [9]. While these systems contribute meaningfully to health tracking and psychological well-being, few have addressed the physical interaction challenges associated with daily dressing routines. Specifically, the usability of wearable systems under physical constraints – such as the need for one-handed operation – remains underexplored. This study addresses this gap by focusing on the development of a maternity support garment that enhances ease of use through one-handed donning and doffing, without compromising support functionality or comfort.

To effectively support pregnant individuals—particularly those in the final trimester—as well as new mothers, maternity support garments must address both ergonomic functionality and ease of use. One of the key challenges in existing designs is the difficulty in donning and doffing. This study aims to develop a wearable solution that integrates effective support features with a one-handed operability, offering improved usability without compromising comfort or functionality.

2 Device Design

2.1 Design Consideration & Criteria

In the design of a maternity support band that enables one-handed operation, it is critical to ensure that both the closure mechanism and the tightness adjustment system are specifically engineered to function effectively with a single hand, without compromising usability or support. Regarding the closure mechanism, it is essential not only that it can be operated with one hand, but also that it requires minimal physical effort. Pregnant individuals experiencing upper extremity pain or reduced muscle strength may encounter difficulties manipulating conventional fasteners, such as Velcro. Therefore, closures that demand low force and support one-handed operation are particularly important. Examples of suitable systems include BOA®, one-handed zipper systems with pre-threaded tracks, and magnetic clasps—all of which facilitate effective one-handed use while minimizing physical strain.

Furthermore, to enable true one-handed operation, support garments must first be capable of temporarily self-anchoring to the body. This initial fixation reduces the need for bilateral coordination and allows users to complete the tightening or adjustment process with a single hand. SMA springs present a promising solution for achieving such preliminary anchoring. SMA springs, typically made from Nitinol—a nickel-titanium alloy—are smart materials capable of returning to their original shape after deformation when exposed to heat. This thermally activated behavior allows them to function as compact actuators,

making them suitable for integration into wearable systems that require hands-free or assistive tightening mechanisms [10].

In addition to the closure mechanism, adjustability is a critical factor in maintaining the garment’s effectiveness throughout the dynamic changes of pregnancy. As the body undergoes progressive expansion, particularly in the abdominal and pelvic regions, the support garment must accommodate fluctuations in size and shape without compromising comfort or compression. Systems that allow adjustments are especially valuable, enabling users to tailor the level of support in response to daily physical demands or varying postures. Building on these principles, the present study introduces SoloBand—an integrated design solution that addresses these functional challenges through both mechanical and material innovations.

2.2 Device Development & Prototypes

2.2.1 Early Prototypes and Iterations Compression, efficiency, and support served as the core design principles guiding the development of the SoloBand system. To address the functional needs of one-handed operation without compromising secure fit or adaptability, two advanced technologies were integrated into the design: the BOA® closure system and SMA springs.

The BOA® system (Figure 1), commonly used in performance footwear like ski boots, was chosen for its precision and ease of use. It features a push-to-engage dial that tightens high-tensile cords between garment layers with a simple twist, providing uniform compression with minimal effort. Releasing the system is equally intuitive—popping the dial outward instantly loosens the cords. This one-handed, user-friendly mechanism is well-suited for pregnant individuals with limited upper limb mobility and new mothers with one hand often occupied. In parallel, SMA springs were incorporated into the design as an assistive mechanism for automated tightening. As shown in Figure 1, fifteen thermally responsive SMA coils are positioned along the posterior side of the garment and secured with snaps. These coil-shaped SMA wires contract when exposed to heat, causing them to shorten and generate tension across the band. Importantly, the actuation of the SMA wires serves as an initial self-anchoring mechanism, allowing the maternity support band to achieve initial fixation on the body. Once this preliminary attachment is established, users can then utilize the BOA® system to perform secure tightening, enabling full adjustment with one hand and improving both ease of use and compression level.

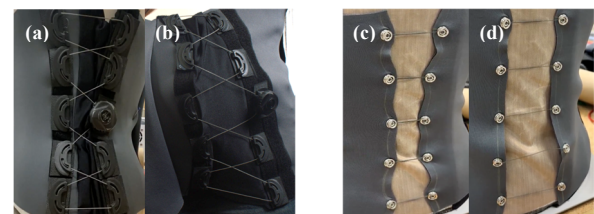


Figure 1: BOA® system tightened (a) and loosened (b), and SMA coils tightened (c) and loosened (d).

The selected fabrics for the prototype included neoprene and a nylon-elastane blend. The choice of nylon and elastane was

informed by a market analysis of existing maternity support bands, which revealed that this combination is commonly used due to its flexibility and comfort [11]. The nylon-elastane prototype provided high elasticity but lacked structural integrity for adequate support and compression, revealing the need for reinforcement materials like neoprene to meet design goals.

2.2.2 Final Design To enhance overall comfort and adaptability, the SoloBand design incorporated structural and material features beyond compression mechanisms. An optional lower support strap was added at the bottom of the band to provide targeted abdominal lift during the third trimester. This strap is fully detachable and can be removed during the postpartum period to improve comfort and allow for greater mobility (Figure 2). In terms of safety and material protection, a layer of poly-coated kraft paper (Teflon) was embedded along the inner surface of the posterior panel, serving as a thermal barrier between the SMA coils and the user's skin. This protective layer mitigates the risk of burns or overheating during SMA activation. Additionally,



Figure 2: Final prototype (front view).

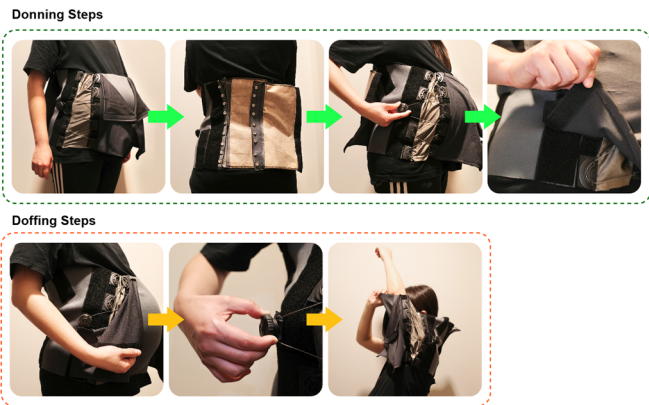
a discreet flap was incorporated to cover the SMA components post-activation, enhancing both aesthetics and mechanical durability (Figure 3).



Figure 3: Final prototype (back view).

From a functional standpoint, the SoloBand's one-handed operability was approached through a two-phase design strategy. The first phase focuses on initial positioning, achieved through SMA spring contraction upon thermal activation. As the SMA coils contract, they gently draw the band into alignment around the torso, anchoring the garment without the need for bilateral hand use. At this stage, the primary objective is secure placement rather than compression. In the second phase, the BOA® system is engaged to fine-tune tension. With a single-hand twist of the dial, users can evenly tighten the band, delivering controlled compression and lumbar support tailored to individual needs. This two-phase approach balances ease of use, adjustability, and support. Figure 4 shows the donning/doffing steps.

Figure 4: Donning: Wear SoloBand around the abdomen-



SMA actuates for initial positioning-Adjust BOA dial- Attach the cover panel; Doffing: Detach the top cover panel-Loosen BOA dials on both sides-Remove SoloBand.

3 Preliminary Results

To assess the feasibility and functional performance of the SoloBand prototype, this study conducted a series of preliminary evaluations aligned with the key design criteria. The evaluation was divided into three domains: one-handed donning and doffing efficiency, support capability, and compression performance. All tests were performed using one subject and repeated across three trials to ensure consistency and gather comparative data.

3.1 One-Handed Donning & Doffing Efficiency

To evaluate the usability of the two-phase tightening mechanism, we recruited three non-pregnant female participants (average age: 24.7 years) and measured the time required to don and doff the band. Participants were asked to wear the band using the SMA-based initial anchoring followed by BOA®-based compression tightening. On average, participants achieved full securement within 43 seconds, while removal—including BOA® release and band removal—took approximately 18 seconds (Table 1).

Table 1: Time of donning and doffing the SoloBand.

Trial	1	2	3	Average
Donning (sec)	35.7	58.2	36.2	43.4
Doffing (sec)	11.7	23.4	20.2	18.4

3.2 Support Capability

To preliminarily evaluate the SoloBand's ability to provide abdominal support, we simulated a late-pregnancy condition using a mock pregnancy belly. To capture the displacement, we inserted two pins—one before and one after applying the band—then read the vertical shift. The test was repeated three times. The results showed that the mock pregnancy belly was lifted (Table 2).

Table 2: Lifting distance of the SoloBand.

Trial	Displacements (inch)
1	0.65
2	0.62
3	0.50
Average	0.59

3.3 Compression Performance

To quantify band's ability to deliver circumferential compression, we fitted the prototype onto a third-trimester pregnancy mannequin and employed force-sensitive resistors (FSR Model 406) to capture applied pressure. Sensors were placed at four locations: lower abdomen, left and right lateral abdomen, and lower back. After donning the band, the BOA® system was tightened to its maximum tension. This procedure was repeated across three trials. The results indicated that the lower abdomen received the lowest average pressure, while the left and right lateral regions recorded symmetrical forces. The lower back pad exhibited the highest average force, suggesting effective localized compression (Table 3).

Table 3: Compression force on different regions of the torso.

Trial	Abdomen		Back	
	Lower	Lateral (L)	Lateral(R)	Lower back
1	104	234	216	203
2	100	227	237	298
3	103	229	224	214
Average	102.3	230	225.7	238.3

4 Discussion

The results indicate that the SoloBand is feasible for one-handed donning and doffing, achieving efficient operation within a short time frame, consistent with the design goal of ease of use. Regarding support functionality, the average uplift suggests a measurable unloading effect on the lower abdomen, which could help reduce lumbar strain. However, the mock pregnancy belly used in this evaluation weighed considerably less than a typical third-trimester load (approximately 5–6 kg, including fetus, placenta, amniotic fluid, and uterus) [1]. Thus, while the band's mechanical potential is evident, future validation with

appropriately weighted mock bellies or through human testing is needed.

There is limited research on the optimal compression force for maternity garments. One study found that a pelvic belt with 50 N (≈ 5.1 kg) of tension was effective in reducing sacroiliac joint mobility and relieving pelvic pain during ADLs [12]. In this study, the SoloBand showed measurable compression at the abdomen and lower back, but values were relatively low compared to clinical references. This discrepancy may be due to factors such as insufficient material stiffness, a low reduction factor (RF), and the limited sensitivity of the FSRs used [13, 14]. Despite these limitations, the SoloBand may still offer low to moderate compression levels that can provide meaningful support for everyday use.

Three key limitations point to future improvements. First, using a heat gun for SMA activation is unsafe and impractical; future versions can employ body-temperature-responsive SMA materials that activate with natural heat. Second, the prototype lacks size adaptability, underscoring the need for pregnancy-specific anthropometric data and standardized sizing. Third, to enhance the efficient and safe SMA activation, we propose integrating an embedded electrical circuit which connects directly to SMA elements with a switch-based interface. Thus, the system could be electronically programmed to reach the temperature needed for the SMA for shape transformation and actuation.

Looking beyond the current prototype, the next-generation vision for the SoloBand will incorporate computing and automation to enable a more intelligent and potentially hands-free user experience. By embedding microcontrollers, sensors, and actuators directly into the garment, the upgraded SoloBand will be able to autonomously detect changes in body posture, abdominal load, or user movement patterns and adjust its tension accordingly. This vision will reimagine the SoloBand as a smart wearable system that not only supports the body, but also actively assists in dynamic, everyday activities—empowering users with greater independence and convenience.

5 Conclusion

Preliminary findings suggest that the SoloBand is a feasible one-handed solution that improves maternity garment usability. Its two-phase system—SMA springs for self-anchoring and a BOA® dial for compression—addresses key challenges in independent donning and doffing. Future work will optimize support through body mapping, automate SMA activation, and incorporate user feedback to refine SoloBand into a practical, inclusive, and ergonomic daily support solution.

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